

Power Series Solutions to Nonlinear Ordinary Differential Equations

MATH 689 Homework 4

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February 17, 2026

- Rewrite as a complex function and expansion.
- Identify the expansion point z_0 .
- Identify all of the singularities $z_{s1}, z_{s2}, \dots$.
- Identify the closest singularity (or singularities) to z_0 , and label it/them $z_{s,*}$.
- Find the radius of convergence, R , using the corollary of Taylor's theorem.
- Make a sketch of the complex plane, showing the points z_0 and $z_{s1}, z_{s2}, \dots$ as well as the circle of convergence centered at z_0 with radius R and with the closest singularity, $z_{s,*}$, lying on the circle.

1. Homework 3 Problem 1 revisited

$$f(x) = \frac{e^x}{1+x} = \sum_{n=0}^{\infty} c_n x^n, \quad |x| < R \quad (1)$$

Rewriting $f(x)$ as a complex function and expansion

$$f(z) = \frac{e^z}{1+z} = \sum_{0 \leq n \leq \infty} \left(\frac{1}{n!} - c_{n-1} \right) (z - z_0)^n, \quad c_0 = 1, \quad n \geq 1, \quad |z - z_0| < R \quad (2)$$

Notice that $z_0 = 0$, with only one singularity $z_{s1} = -1$ because $f(z_{s1})$ is non-analytic. It follows that the closest singularity to z_0 is $z_{s1} = -1$, so $z_{s,*} = -1$. Then using corollary of Taylor's theorem we have the radius convergence as $R = |z_{s,*} - z_0| = |-1 - 0| = 1$

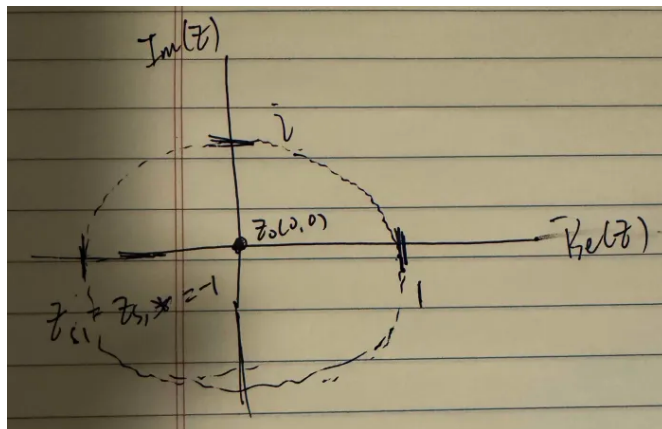


Figure 1: Sketch of R on $\mathbb{C}$

2.

$$f(x) = \frac{\ln(x-3)}{x^2 - 2x + 2} = \sum_{n=0}^{\infty} a_n x^n, \quad |x| < R \quad (3)$$

Rewriting $f(x)$ as a complex function and expansion

$$f(z) = \frac{\ln(z-3)}{z^2 - 2z + 2} = \sum_{0 \leq n \leq \infty} a_n (z - z_0)^n, \quad |z - z_0| < R \quad (4)$$

Here we expand about $z_0 = 0$, then from the numerator $\ln(x-3)$, there is a singularity $z_{s1} = 3$, then from the denominator of $f(z)$ there is two singularities $z_{s2} = 1+i$, $z_{s3} = 1-i$. From calculating the radius of convergence using Taylor's theorem

$$\begin{aligned} |z_{s1} - z_0| &= |3 - 0| = 3 \\ |z_{s2} - z_0| &= |(1+i) - 0| = \sqrt{2} \\ |z_{s3} - z_0| &= |(1-i) - 0| = \sqrt{2} \end{aligned} \quad (5)$$

So $z_{s,*} = z_{s1,2} = \{1 \pm i\}$ with $R = \sqrt{2}$.

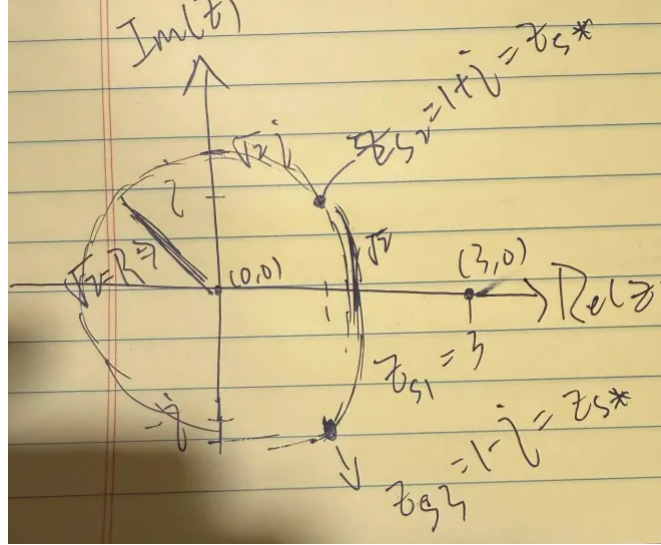


Figure 2: Rough Sketch on $\mathbb{C}$

3.

$$f(x) = \frac{x^{3/2}}{x^2 + 1} = \sum_{n=0}^{\infty} a_n (x-1)^n, \quad |x-1| < R \quad (6)$$

If we expand $f(z)$ about $z_0 = 1$ we will get a singularities point at $x = 0$ for second derivatives or higher.

$$f(z) = \frac{z^{3/2}}{z^2 + 1} = \sum_{0 \leq n \leq \infty} a_n (z - z_0)^n, \quad |z - z_0| < R \quad (7)$$

Explicitly we have the first order derivative as

$$f'(z) = \frac{3/2 \sqrt{z}(z^2 + 1) - z^{3/2}(2z)}{(z^2 + 1)^2} \quad (8)$$

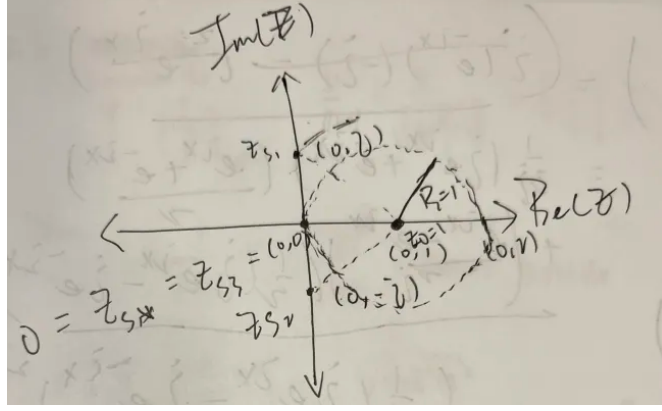


Figure 3: Rough Sketch on $\mathbb{C}$

Notice if we take the second derivative from here the terms like $\sqrt{z}, z^{3/2}$ will have their power go to negative for higher derivatives thus being non-analytic at $z = 0$. Then we say $z_{s1,2} = \pm i, z_{s3} = 0$. Then computing the radius of convergence yields

$$\begin{aligned} |z_{s1} - z_0| &= |i - 0| = 1 \\ |z_{s2} - z_0| &= |-i - 0| = 1 \\ |z_{s3} - z_0| &= |0 - 0| = 0 \end{aligned} \quad (9)$$

with $z_{s,*} = z_{s3} = 0$ and radius $R = 1$.

5.

$$f(x) = \tan(x) = \sum_{n=0}^{\infty} a_n x^n, \quad |x| < R \quad (10)$$

Writing $f(x)$ as a complex function and its expansion with B_j as the Bernoulli's numbers yields

$$f(z) = \tan(z) = \sum_{1 \leq n \leq \infty} \frac{B_{2n}(-4)^n(1-4^n)}{(2n)!} z^{2n-1}, \quad |z - z_0| < R \quad (11)$$

With the power series being well defined above we know that the function at the n derivative evaluated at $z_0 = 0$ will be analytic. So the key to finding radius must lie in the zeroth derivative namely we have

$$\tan(z) = \frac{\sin(z)}{\cos(z)} \quad (12)$$

Notice that at $z = \frac{\pi}{2} + n\pi, \forall n \in \mathbb{Z}$, we have a singularity for $f(z)$ since $\cos(\frac{\pi}{2} + n\pi) = 0$. If we look at the modulus between z_0 to all $z_{s,i}$, for $\text{card}(\{i\}) = \text{card}(\mathbb{Z})$, since the radius is the closest value to the $z_0 = 0$

$$R = \min_n \left(\left| \frac{\pi}{2} + n\pi - 0 \right| \right) \quad (13)$$

From this we get that $R = \pi/2$ from $\arg \min_n (|\frac{\pi}{2} + n\pi - 0|) = n = \{0, -1\}$, so $z_{s1} = \pi/2, z_{s2} = -\pi/2$, and $z_{s,*} = \{\pm\pi/2\}$.

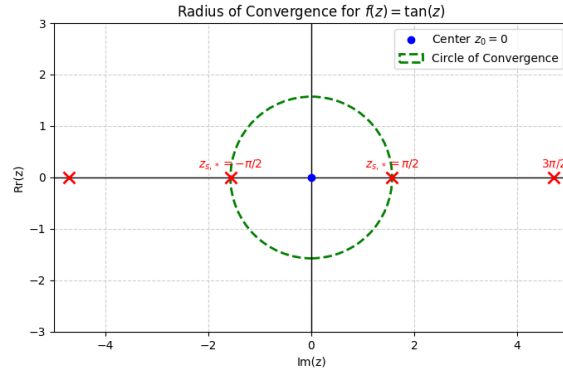


Figure 4: Radius of Convergence for $f(z) = \tan(z)$

6.

$$f(x) = \tan(x) = \sum_{n=0}^{\infty} a_n (x-4)^n, \quad |x-4| < R \quad (14)$$

Rewriting as a complex function we obtain

$$f(z) = \tan(z) = \sum_{1 \leq n \leq \infty} \frac{B_{2n}(-4)^n(1-4^n)}{(2n)!} (z-4)^{2n-1}, \quad |z-4| < R \quad (15)$$

To find the radius of convergence R , we identify the closest singularity $z_{s,*}$ to the center $z_0 = 4$. The singularities of $\tan(x)$ occur at $z_s = \frac{\pi}{2} + n\pi$ for $n \in \mathbb{Z}$ as follows from question (5).

We calculate the modulus $|z_{s,i} - z_0|$ for the smallest radius of convergence

$$R = \min_n \left(\left| \left(\frac{\pi}{2} + n\pi \right) - 4 \right| \right) \quad (16)$$

We find that $R = \frac{3\pi}{2} - 4$ with $n = 1$, so $z_{s1} = z_{s,*} = \frac{\pi}{2} + \pi$.

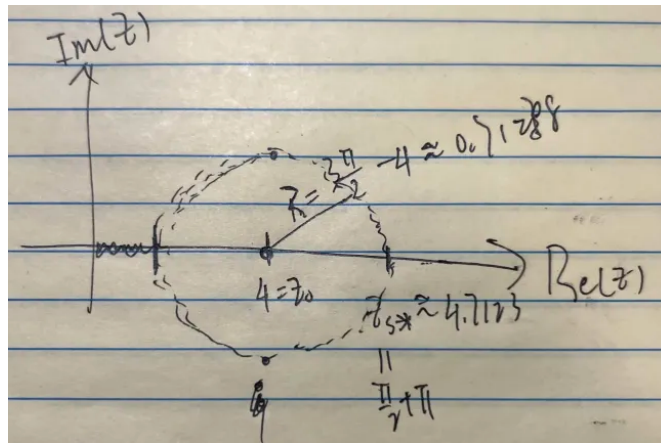


Figure 5: Rough Sketch on $\mathbb{C}$

4. Choose the problem for your midterm project

(a) Find a nonlinear ODE that you would like to solve. It will need to be one whose analytical solution is not known (approval required). You may consult the internet, the library, or books. Look for ODEs in numerical methods or dynamical systems chapters or books.

A mathematical model for chronic myelogenous leukemia (CML) and T cell interaction[1]

The model we will be solving is a adjusted version of the original system of nonlinear ODEs in Moore and Li's paper from Anne Talkington, Claudia Dantoin, and Rick Durrett, Duke U. [2]

$$\begin{aligned}\frac{dT}{dt} &= aT(1 - bT) - \gamma_c ET \\ \frac{dE}{dt} &= s_e - d_e E + \alpha_n k_n N \frac{T}{T + g} - \gamma_e ET \\ \frac{dN}{dt} &= s_n - d_n N - k_n N \frac{T}{T + g}\end{aligned}\tag{17}$$

With initial conditions from the **Tumor Free Equilibrium** theorem we use $T(0) = 0, E(0) = s_e/d_e = 600, N(0) = s_n/d_n = 1080$ and is stable when $a < \gamma_c E(0)$

The units of T, E , and N in units of *cells/ μ l* with coefficients above in (17) known as

$$b = 1/30000, \quad \gamma_c = 0.005, \quad \gamma_e = 0.0001, \quad g = 100$$

$$k_n = 0.001, \quad \alpha_n = 100, \quad s_e = 36, \quad d_e = 0.06, \quad s_n = 43.2, \quad d_n = 0.04$$

(b) You must have it approved before 2/17. When considering your ODE, note that the midterm project will require you to: *Approved through email and in person discussion*

- Find the numerical solution,
- Find the power series solution,
- Plot them on the same graph,
- Use the tools learned in class to estimate the radius of convergence.

References

- [1] Helen Moore and Natasha K Li. A mathematical model for chronic myelogenous leukemia (cml) and t cell interaction. *Journal of theoretical biology*, 227(4):513–523, 2004.
- [2] Anne Talkington, Claudia Dantoin, and Rick Durrett. Ordinary differential equation models for adoptive immunotherapy. *Bulletin of mathematical biology*, 80(5):1059–1083, 2018.